Stripe Meets Window: Merging Stripe and Windowed Mamba Scans with Learned Routing for Local and Global Context in Remote Sensing Super-Resolution

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Abstract

We study how the order in which a Mamba network reads image tokens affects $\times 4$ super-resolution (SR) of aerial and satellite images. We start from EAMamba, which splits the channels into groups that are scanned along fixed directions. We replace the reorder step of one group with (i) a stripe scan traversed in the nested S-shaped (NSS) manner, and (ii) a concentric-ring scan applied inside windows, and we add (iii) a small router network that learns which of these scans to use for each window. The window scans keep local detail, the stripe scan carries context across window borders, and the remaining direction scans keep global context, so the network merges local and global reading. All variants share EAMamba's architecture and training recipe, so differences come from the scan. We train on AID with bicubic downsampling and test on AID, LoveDA-Val, DOTA-Test and DIOR. We compare with MambaIR and SwinMamba under identical data and training. Because recent gains in this field are a few tenths of a dB, we report several seeds, ordering controls and per-category results. If no variant wins, the project still delivers a controlled study of scan order for aerial SR.

1 Introduction

Aerial and satellite images are large, contain many small repeated structures (cars, roofs, storage tanks, runway markings) and lose most of their fine detail when downsampled. Mamba-based restoration networks are attractive here because their cost grows linearly with the number of tokens, but a Mamba layer reads a one-dimensional sequence, so the order in which the two-dimensional feature map is flattened decides which tokens are neighbours in the sequence and which are far apart. Neighbours in the image can end up far apart in the sequence, and Mamba's state fades with distance (the "local pixel forgetting" problem described in MambaIR).

Existing work fixes this by adding more scan directions (EAMamba, VMamba-style cross scans), by adding convolutional branches (MambaIR), or by designing new traversal paths (MaIR's nested S-shaped scan, a stripe scan for hyperspectral SR, a ring scan for classification). These ideas have been evaluated on different tasks and datasets, so it is hard to tell which scan helps aerial SR and when. We propose to compare them inside a single backbone, EAMamba, under one protocol, and to test whether a small network that chooses the scan per window can beat the best fixed scan.

2 Related Work

EAMamba (ICCV 2025) builds a U-shaped restoration network from MambaFormer blocks and splits the channels into groups scanned along different directions (all-around scanning). Its published SR results are on RealSR, so for aerial SR it must be retrained. **MambaIR** (ECCV 2024) adds local convolution and channel attention to Mamba for restoration; MambaIRv2 (CVPR 2025) adds attentive state-space modelling; **MaIR** (CVPR 2025) introduces the nested S-shaped scan and a block that merges scan sequences, arguing that a scan should preserve locality and continuity.

Stripe and ring scans. A stripe-scan variant of HSRMamba (arXiv 2025, hyperspectral SR) introduces cross-window interaction through stripes; in its own ablation on one dataset a window-only scan was worse than a global scan, and the stripe scan was best. PRISMamba (ICML 2026) scans concentric rings for rotation robustness in classification. **Adaptive scans**: LocalMamba searches scan directions per layer, VAMamba plans a path from importance scores, and D2Mamba finds paths with A* search.

Windowed and local scans. LocalMamba (arXiv 2024) scans inside local windows and searches the scan direction per layer for classification and detection. In remote sensing, SwinMamba (Zhu et al., 2025) uses shifted windows with four-direction scans in its early stages and global scans in its late stages for semantic segmentation; AtrousMamba (2025) groups tokens into dilated windows for change detection; and HSRMamba uses windowed Mamba for hyperspectral SR [8]. An experimental study on LoveDA, Potsdam and Vaihingen compared 22 scan strategies for segmentation and found that no single strategy wins (Zhu et al., 2024), which motivates choosing the scan per window rather than fixing it. None of these tests ring paths inside windows, or compares scans for RGB aerial SR; we have not searched exhaustively.

Aerial SR. Rep-Mamba, LAMA-SR and DFSMamba are Mamba-based aerial SR models. TexADiff (CVPR 2026) is a diffusion model for aerial SR that uses the Real-ESRGAN degradation pipeline on LoveDA, DOTA, AID and RSC11; we borrow only its choice of test subsets, and keep its degradation for future work. We did not find a study that compares stripe, NSS, ring and learned scan selection inside one restoration backbone for aerial SR; we have not searched exhaustively.

3 Approach

3.1 EAMamba recap

EAMamba is built from MambaFormer blocks. Each block mixes tokens with a multi-head selective scan (MHSSM), then refines channels with an MLP:

$$X' = X + \mathrm{MHSSM}\big(\mathrm{LN}(X)\big), \qquad X'' = X' + \mathrm{MLP}\big(\mathrm{LN}(X')\big). \tag{1}$$

Inside the scan, the channels are split into groups, and each group i is reordered by T_i, scanned by a state-space model, and restored by T_i^{-1}:

$$\mathrm{MHSS}(X) = \mathrm{Concat}_i\Big(T_i^{-1}\big(\mathrm{SSM}(T_i(X_i))\big)\Big), \tag{2}$$

where T_i is a direction: horizontal, vertical, diagonal or flipped diagonal, each also reversed. Our changes only replace T_j for one group j (and later for more groups), so the parameter count and the other groups stay the same.

3.2 Candidate scans

Every candidate is a permutation of the tokens, so it is implemented as an index array computed once per input size.

Stripe scan with NSS traversal (E1). The feature map is cut into stripes of width s and the tokens inside each stripe are visited in the nested S-shaped manner of MaIR, so neighbouring tokens stay neighbours in the sequence and tokens across a window edge become close. We test $s \in \{2, 4, 8\}$, horizontal and vertical stripes, and one group versus all groups.

Ring scan inside windows (E2). The map is split into $w \times w$ windows (shifted by $w/2$ in alternate blocks to avoid boundary artefacts). Inside a window, a token at (i, j) has ring index

$$r(i, j) = \min(i,\, j,\, w - 1 - i,\, w - 1 - j), \tag{3}$$

and tokens are ordered by r, then clockwise along the ring, either outside-in or inside-out. Windows are scanned one after another, and the other groups keep their global direction scans so that long-range context is not lost.

3.3 Learned scan router (E3)

Let $\mathcal{T} = \{T^{\text{dir}}, T^{\text{NSS}}, T^{\text{ring}}\}$ be the $K = 3$ candidate scans for the routed group. For a window with features $x \in \mathbb{R}^{N \times C}$, a router computes

$$g = \tfrac{1}{N} \textstyle\sum_n x_n \in \mathbb{R}^C, \qquad \pi = \mathrm{softmax}(W g/\tau) \in \mathbb{R}^K, \qquad y = \textstyle\sum_k \pi_k \, T_k^{-1}\big(\mathrm{SSM}(T_k(x))\big). \tag{4}$$

No scan label is needed: the router is trained end to end with the usual L1 reconstruction loss. The softmax output π is a vector of K weights that sum to one, one weight per candidate scan, and it is the quantity we study. During training the model uses all K scans as a soft mixture, so π_k is the share of scan k in the output. Because the weights are learned from the reconstruction loss, a larger π_k means the network found scan k more useful for that layer or window. We log π and report its mean per layer, per scene category and per test set, which tells us which scan helps most and where. Two inference modes are compared: the full soft mixture (best quality, K times the scan cost) and top-1, where only the scan with the largest weight runs for each window, grouping windows by chosen scan so they can be batched (cheaper). A small balance term discourages collapse onto one scan, and an optional entropy term sharpens π so the weights are easier to read; Gumbel-softmax with a straight-through estimator is a further variant. Router granularity is a variant: (a) one learned weight vector per layer (static, similar to a scan-direction search), (b) one choice per image, (c) one choice per window. Variant (a) is the non-adaptive control for (c). The soft mixture costs K times the scan compute of one group, which we report.

3.4 Controls (E4)

To tell whether any gain comes from the scan and not from extra capacity or luck, we run: a random token order in the replaced group, the uniform average of the three scans, the best single fixed scan, and the router with its decision shuffled across windows.

4 Experimental Plan

4.1 Data and degradation

The task is $\times 4$ SR with bicubic degradation, the standard setting of aerial SR papers, so every method sees identical LR/HR pairs.

Table 1: Datasets. Test subsets follow TexADiff where it gives them; the AID split and the DIOR subset are our own, with fixed seeds and released file lists.

Dataset	Role	Size used	Notes
AID	train / test	3000 train, 900 test, 640×640	our fixed random split; baseline training set
LoveDA-Val	cross-dataset test	full validation set	as in TexADiff; own runs only
DOTA-Test	cross-dataset test	1000 random 512×512 crops	as in TexADiff; images only
DIOR	cross-dataset test	1000 random images	our subset; boxes allow object metrics
SIRI-WHU	optional real-image test	full set (2 m)	no degradation applied; no reference

Degradation. Each high-resolution (HR) image is downsampled by four with bicubic interpolation with anti-aliasing to give the low-resolution (LR) input. Test LR images are generated once, saved as PNG files and shared by all methods; training LR patches are cut from the HR crops in the same way. We fix and document the exact operator (library and anti-aliasing setting).

Later investigations. Once the bicubic baseline is solid, we plan to repeat the best scan variants under harder degradations: the Real-ESRGAN pipeline as used by TexADiff (random blur, resizing, noise and JPEG compression applied in two rounds), heavier noise or blur, and a sensor-style blur, to see whether a scan advantage grows when the input is poorer. These are out of scope for the main comparison.

4.2 Baselines and training

EAMamba (ICCV 2025) retrained on AID is the main baseline. We compare with **MambaIR** (ECCV 2024) and **SwinMamba**, by which we mean Swin-UMamba (Mamba-based UNet with Swin-

style windows; arXiv:2402.03302), retrained on AID with the same data, degradation and schedule. Swin-UMamba was designed for medical image segmentation, so we replace its segmentation head with a $\times 4$ pixel-shuffle upsampling head and train it from scratch, without ImageNet pretraining, as for all other models. We use a standard recipe: Adam (β_1=0.9, β_2=0.999), L1 loss, learning rate 1×10^{-4} halved every 200 epochs, about 500 epochs, mini-batches of four patches, random flips and rotations. If the default EAMamba configuration does not fit one 24 GB GPU at this patch size, we use a reduced width and depth and apply it to every variant. Headline comparisons use three seeds and report mean $\pm$ std.

4.3 Metrics and analysis

- PSNR and SSIM on the Y channel on every test set. For the baselines (EAMamba, MambaIR, SwinMamba) we monitor only PSNR, SSIM, FLOPs and model size.
- Per-category PSNR and SSIM on the 30 AID scene classes, to see where scans differ.
- Cost: model size (parameters) and FLOPs at a fixed LR input of $1 \times 3 \times 128 \times 128$ for all methods, computed with the same tool.
- Scan weights: the mean router weight vector π per layer, per scene category and per test set, and a histogram of top-1 choices.
- Object regions: PSNR inside annotated small and large boxes against outside boxes, on annotated DIOR images (and DOTA train/val images if used). DOTA-Test annotations are withheld, to our knowledge; PSNR needs only the images, so DOTA-Test is used for PSNR/SSIM only. Confirm before running.
- Paired per-image differences with bootstrap confidence intervals between each variant and EAMamba.

4.4 Experiments

Table 2: Planned experiments. Each variant changes only the scan of the listed group.

ID	Experiment	What varies
E0	EAMamba, MambaIR, SwinMamba on AID	baselines, 3 seeds
E1	Stripe scan with NSS traversal	stripe width, orientation, one group vs. all groups
E2	Ring scan in windows	window $\{8, 16\}$, outside-in/inside-out, shifted windows
E3	Learned router over {dir, NSS, ring}	per layer / per image / per window
E4	Controls	random order, uniform mix, best fixed scan, shuffled router
E5	Best variants, cross-dataset	LoveDA-Val, DOTA-Test, DIOR (trained on AID only)

Hypotheses. H1: stripe/NSS scans match or beat the default direction scan on local detail. H2: ring scans inside windows help small structures. H3: the router chooses different scans for different scene types and beats the best fixed scan. H4: gains are largest on structured scenes (parking lots, storage tanks, ports).

Risks. Expected gains are about 0.1 dB and may sit inside seed noise; we answer with three seeds, paired bootstrap intervals and controls. The router may collapse to one scan; the balance term and the per-layer control address this. Soft routing multiplies scan cost by K; we report FLOPs at inference with top-1 routing. Windowed scans may lose global context, which is why the other groups stay global.

4.5 Timeline (15 weeks)

5 Conclusion

We propose a controlled study of scan order inside EAMamba for aerial super-resolution: stripe/NSS and window-ring scans, and a router that learns which scan to use, evaluated under bicubic degradation. If one variant beats the baseline beyond seed noise, that is the contribution; if not, we deliver a reproducible comparison with controls and per-category analysis. Later extensions, once the

Weeks	Work
1–2	Environment, data splits, bicubic degradation and fixed test LR sets, first EAMamba run on AID
3–4	Full baselines (EAMamba, MambaIR, SwinMamba), seeds, evaluation scripts
5–6	E1 stripe with NSS traversal
7–8	E2 ring scan in windows
9–11	E3 router and E4 controls
12	E5 cross-dataset tests
13	Per-category, scan-choice and object-region analysis
14–15	Writing and buffer

baseline is solid, include harder degradations (Real-ESRGAN-style and others), object-aware window selection, and similarity-based ordering of tokens.

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